

Anuj Kankani

PhD Candidate in Physics, West Virginia University · Expected graduation May 2027

RESEARCH · Gravitational Waveform Modeling · Numerical Relativity · General Relativistic Particle-in-Cell (GRPIC) Simulations

EDUCATION

- 2021–27** **West Virginia University** — PhD in Physics
expected Expected graduation May 2027 · Advisor: Sean T. McWilliams · GPA 3.9/4.0 · Morgantown, WV
- 2017–21** **Texas A&M University** — BS in Physics (Honors)
Minors in Computer Science and Astrophysics · GPA 3.84/4.0 · College Station, TX

PUBLICATIONS

- The Direct Wave is Not a Meaningful Test of Horizon Properties
Anuj Kankani and Sean T. McWilliams · submitted to *Physical Review Letters* · arXiv:2607.02380
- gwBOB: A Python Package for Analytical Merger-Ringdown Gravitational Waveforms
Anuj Kankani, Angel Morales, Suchindram Dasgupta and Sean T. McWilliams · submitted to *Journal of Open Source Software*
- Direct Waves in Black Hole Binary Mergers: Insights from the Backward One-Body Model
Anuj Kankani and Sean T. McWilliams · Phys. Rev. D **114**, 084025 (2026) · arXiv:2603.15474
- Analyzing Analytical Black-Hole Binary Merger Waveforms with the Backwards One-Body Formalism
Anuj Kankani and Sean T. McWilliams · Phys. Rev. D **114**, 084007 (2026) · arXiv:2510.25012
- Modeling Relative Peak Times of Gravitational Wave Harmonics
Anuj Kankani and Sean T. McWilliams · Phys. Rev. D **112**, 124051 (2025) · arXiv:2506.17492
- Testing the Boundary-to-Bound Correspondence with Numerical Relativity
Anuj Kankani and Sean T. McWilliams · Phys. Rev. D **110**, 064033 (2024) · arXiv:2404.03607
- Optimization and benchmarking of the thermal cycling algorithm
A. Barzegar, **A. Kankani**, S. Mandrà & H. G. Katzgraber · Phys. Rev. E **104**(3), 035302 (2021)

SOFTWARE

- PRIMARY DEVELOPER** **gwBOB** github.com/AnujKankani/BackwardsOneBody
Open-source, user-friendly Python package implementing the Backwards One Body waveform model, with tools for analysis and NR comparison.
- CO-DEVELOPER** **LisaWave-JAX** **IN DEVELOPMENT**
JAX-based LISA data-analysis tools for massive black holes and galactic binaries.
- CONTRIBUTOR** **Einstein Toolkit** einstein toolkit.org
Open-source high-performance-computing codebase for numerical relativity.
- CONTRIBUTOR** **Entity** entity-toolkit.github.io/wiki
Kokkos-based General Relativistic Particle-in-Cell code for plasma astrophysics.
- CONTRIBUTOR** **NRPy** github.com/nrpy/nrpy
Python-based symbolic code-generation toolkit for relativistic astrophysics.

CONFERENCES & WORKSHOPS

- 2026 **Plasmas Around Black Holes and Neutron Stars (SCEECs)**
POSTER Driving Temperature Anisotropies in the Jets of Black Holes
- 2026 **16th International LISA Symposium**
CONTRIBUTED TALK BOB the (Waveform) Builder + Direct Waves
- 2026 **APS Global Physics Summit**
CONTRIBUTED TALK BOB the (Waveform) Builder: Optimizing Analytical Black-Hole Binary Merger Waveforms
- 2026 **Center for Gravitational Waves and Cosmology Symposium**
CONTRIBUTED TALK Modeling Extreme Gravity: From Relativistic Plasmas to Gravitational Waves
- 2025 **Prospects in Theoretical Physics — Gravitational Waves from Theory to Observation**
ATTENDED
- 2025 **APS Global Physics Summit**
CONTRIBUTED TALK Merger-Ringdown Modeling of Gravitational Waves Beyond the Dominant Mode
- 2025 **AAS Winter Meeting**
CONTRIBUTED TALK Understanding the Origins of Black Hole Superkicks
- 2024 **North American Einstein Toolkit Workshop**
30-MINUTE TALK Understanding the Origins of Black Hole Superkicks
- 2024 **Penn State Neighborhood Workshop**
CONTRIBUTED TALK Understanding the Origins of Black Hole Superkicks
- 2024 **APS April Meeting**
CONTRIBUTED TALK Testing the Boundary to Bound States Dictionary with Numerical Relativity
- 2023 **AAS Winter Meeting**
POSTER Effect of Baryons on the Dynamical-Friction Timescale of Merging Dark Matter Halos
- 2022 **APS April Meeting**
CONTRIBUTED TALK Effect of Baryons on the Dynamical-Friction Timescale of Merging Dark Matter Halos

AWARDS & FELLOWSHIPS

- 2025 **Mohindar Seehra Research Award**
Departmental award recognizing outstanding research efforts in Physics.
- 2024–2025 **NASA WV Space Grant Graduate Fellowship**
For the proposal “Modeling Black Hole Mergers with a Dynamical Background Spacetime”.
- 2021–2024 **Ruby Distinguished Doctoral Fellow**
Awarded to outstanding doctoral students in STEM fields at West Virginia University.

INVOLVEMENT, SERVICE & OUTREACH

COLLABORATION	LISA Consortium • I am a member of the waveform and astrophysics working group, where I help make sure NR will meet the accuracy requirements demanded by LISA.
DEPARTMENT	Astronomy Journal Club • I organized and led the department’s weekly journal club, facilitating discussions on recent research in astronomy and astrophysics.
OUTREACH	Physics outreach • I give talks to middle- and high-school groups about black holes and gravitational waves, volunteer at physics outreach events, and write articles on astrophysics for local organizations. • I regularly take part in STEM outreach such as Physics Demo Day and the pumpkin drop contest.
SERVICE	Community service • I completed 200 hours of community service as a graduate student, focused on environmental restoration and STEM outreach.